Software Project Estimation

(wideband Delphi estimation technique)

What is Project Estimation?

- **How much will the project take in terms of time, cost, money and manpower?**

- For example, if we are designing a system . Following needs to be answered first,
  - How many developers are needed?
  - How many months will it take?
  - What will be total project cist?
  - What tools or infrastructure is needed?

- After that project manager put forward the proposal for acceptance.

Need for Project Estimation

- Budget planning
- Time scheduling
- Resource allocation
- Risk management
- Stake holders’ communication
- Project feasibility analysis

Need for Project Estimation

Let’s say a team is building an LMS:

<u>Without estimation,</u>

- Deadlines might be unrealistic
- Budget may exceed its limits
- Team might be insufficient

<u>With estimation</u>

- Effort=1200 person-hours (1 person working for an hours)
- Duration = 4months
- Cost= PKR 1,200,000
- Team = 4 developers + 1 tester

Now, planning is structured and controlled.

Estimation techniques

- **COCOMO**
- **Wideband delphi estimation**
- **Functional points estimation**
- **LOC estimation**

For good estimation, following are needed:

- WBS
- deliverables
- A clear effort estimation for each task
- A list of assumptions which were necessary for making the estimate
- Consensus among the project team that the estimate is accurate

Assumptions Make Estimates More Accurate

- Team members make *assumptions* about the work to be done to deal with incomplete information
  - Must estimate the issues needed to make decision
  - **Assumptions must be written down** so that if they prove to be incorrect then it must be back tracked and corrected.

Wideband Delphi

- *Wideband Delphi* is a process that a team can use to generate an estimate
  - The **project manager chooses an estimation team**,

  and gains consensus among that team on the results.
  - Wideband Delphi is a *repeatable* estimation process because it consists of a straightforward set of steps that can be performed the same way each time

Delphi Technique

- Take inputs from multiple stake holders
- Create final estimate on that
- Steps followed:
  - Must have a facilitator/coordinator
  - Create list of experts
  - Create 3-5 questions
    - For example what's better to use for project? Azure / AWS etc.....
  - Reach to all the results
  - Analyze & iterate

The Wideband Delphi Process

- Step 1: Choose the team
  - The project manager selects the estimation team and a moderator. The team should consist of 3 to 7 project team members.
    - The moderator should be familiar with the Delphi process, but should not have a stake in the outcome of the session if possible.
    - If possible, the project manager should not be the moderator because he should ideally be part of the estimation team.

The Wideband Delphi Process

- Step 2: Kickoff Meeting
  - The project manager must make sure that each team member understands the Delphi process, has read the vision and scope document and any other documentation, and is familiar with the project background and needs.
  - The team brainstorms and writes down assumptions.
  - The team has the generated WBS.
  - The team agrees on a unit of estimation.

The Wideband Delphi Process

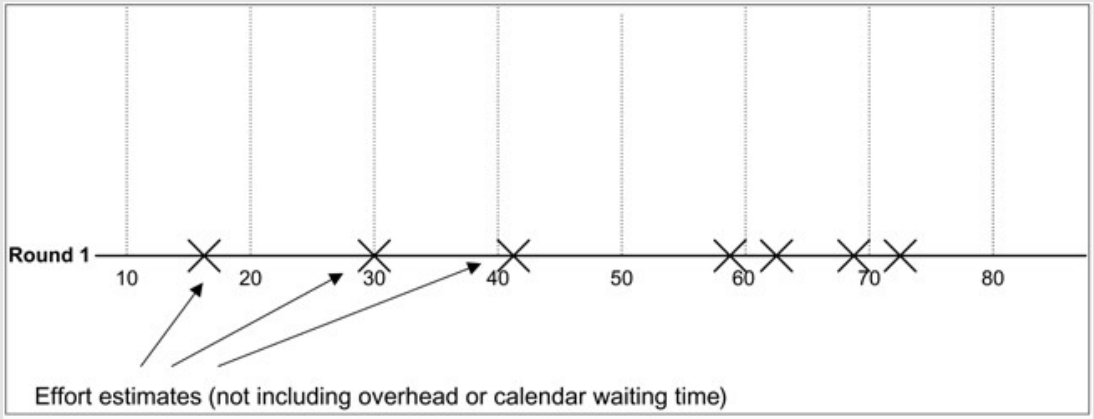
- Step 3: Individual Preparation
  - Each team member independently generates a set of preparation results.
  - For each task, the team member writes down an estimate for the effort required to complete the task,
  - and any additional assumptions he needed to make in order to generate the estimate.

The Wideband Delphi Process

- Step 4: Estimation Session
  - During the estimation session, the team comes to a consensus on the effort required for each task in the WBS.
  - Each team member fills out an estimation form which contains his estimates.
  - The rest of the estimation session is divided into rounds during which each estimation team member revises her estimates based on a group discussion. Individual numbers are not discussed.

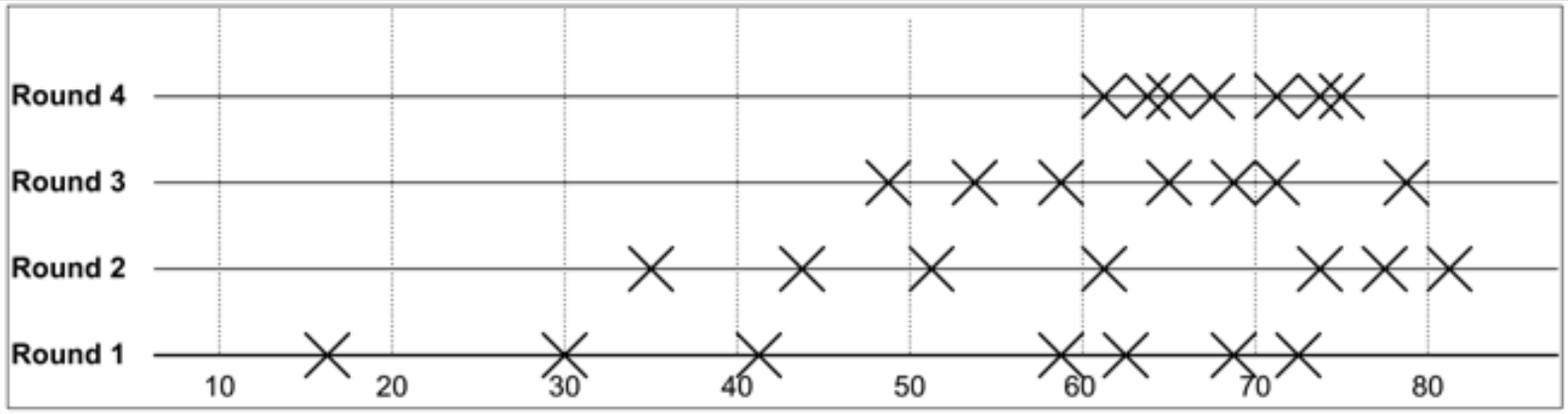
The Wideband Delphi Process

- Step 4: Estimation Session (continued)
  - The moderator collects the estimation forms and plots the sum of the effort from each form on a line:



The Wideband Delphi Process

- Step 4: Estimation Session (continued)
  - The team resolves any issues or disagreements that are brought up.
    - Individual estimate times are not discussed. These disagreements are usually about the tasks themselves. Disagreements are often resolved by adding assumptions.
  - The estimators all revise their individual estimates. The moderator updates the plot with the new total:
  - Goal: to achieve the max Convergence.



The Wideband Delphi Process

- Step 4: Estimation Session (continued):
  - The moderator leads the team through several rounds of estimates to gain consensus on the estimates. The estimation session continues until the estimates converge or the team is unwilling to revise estimates.
  - Delta sheet could also be formed by showing the variation in estimates after a round.
- Step 5: Assemble Tasks
  - The project manager works with the team to collect the estimates from the team members at the end of the meeting and compiles the final task list, estimates and assumptions.
- Step 6: Review Results
  - The project manager reviews the final task list with the estimation team.

Case study

For creating a Chat Application, A list of activities having different tasks are formulated and on the basis of that, project estimation using Wide Band Delphi technique is done. Three estimators took part in the estimation process, shown in Chart given below. Below are the estimation forms deduced from the kickoff meetings. The numbers in the tables depicts the effort estimation in terms of days required for each activity.

Member: Salas	Round 1	Round 2	Round 3
Create UI	3	3	3
Create groups	3	3	4
Send message	6	6	6
Receive message	3	3	3
Add member in groups	5	2	2

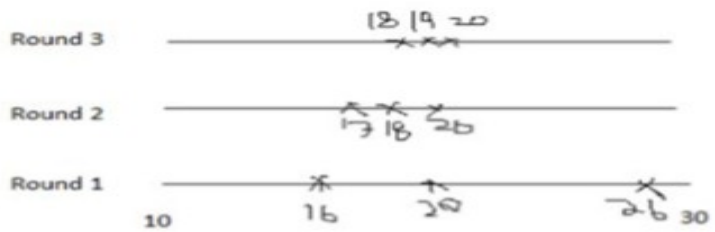
Member: Aliya	Round 1	Round 2	Round 3
Create UI	4	4	4
Create groups	6	5	5
Send message	2	5	5
Receive message	3	3	3
Add member in groups	1	1	2

Member: Ahmed	Round 1	Round 2	Round 3
Create UI	7	7	7
Create groups	8	3	3
Send message	7	6	5
Receive message	2	2	3
Add member in groups	5	2	2

For the above data, do the following:

a) Show how the meeting progressed by drawing a single result graph that represents cumulative estimation for the three members.
b) Create the effort estimation Sheet showing all best, worst and average cases for Round 1 only.

Solution



b)

	Salas	Aliya	Ahmed	Best	Worst	Average
Create UI	3	4	7	3	7	5
Create groups	3	6	8	3	8	6
Send message	6	2	7	2	7	5
Receive message	3	3	2	2	3	3
Add member in group	5	1	5	1	5	3
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